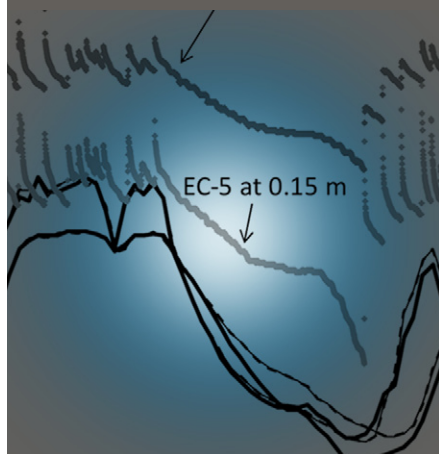


Original Research

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Crop transpiration depends on resistances in the soil–plant–atmosphere system. We present a new deterministic root water uptake model to estimate transpiration and compare it with two other models. We show the sensitivity of actual transpiration to parameters like soil and plant hydraulic properties and root length density distribution with depth.

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Modeling Water Potentials and Flows in the Soil–Plant System Comparing Hydraulic Resistances and Transpiration Reduction Functions

Transpiration reduction functions are often used in hydrological modeling to estimate actual transpiration as a function of soil water status. Empirical reduction functions are most frequently used due to the higher data needs and computational requirements of mechanistic models. Empirical models, however, lack a description of physical mechanisms and their parameters require extensive calibration. We derive a process-based reduction function predicting system potentials, resistances, and water flows. An analytical solution for a special case of Brooks and Corey soils is presented. A numerical version of the reduction function for van Genuchten soils was implemented in the Soil–Water–Atmosphere–Plant (SWAP) hydrological model, allowing predictions for layered soil profiles and root length density variations over depth. The analytical and numerical versions of the model allow an increasingly quantitative insight into the mechanism of root water uptake, such as the existence of a maximum root water uptake rate as a function of soil water status, soil hydraulic properties, root length density, and root radius, in addition to the fact that sensitivity of simulated root water uptake to the radial root conductivity and axial conductance decrease when root length density increases. The approach can be used for the estimation of threshold values for empirical reduction functions.

Abbreviations: CWSI, Crop Water Stress Index; FRF, Feddes et al. (1978) reduction function; JRF, de Jong van Lier et al (2008) reduction function; PRF, proposed reduction function; RMSE, root mean square error; SWAP, Soil–Water–Atmosphere–Plant model.

Modeling of root water uptake and its partitioning over depth is important to support hydrological, meteorological, and crop growth modeling. Under nonlimiting circumstances, root water uptake is at its maximum (potential) rate and transpiration equals potential transpiration T_p . Under limiting conditions, actual transpiration, T_a , is lower than the potential rate T_p . Relative transpiration, T_r , is defined as the ratio between actual and potential transpiration ($T_r = T_a/T_p$). The relation between relative transpiration and soil hydraulic conditions is called the transpiration reduction function (Dekker et al., 2001; Li et al., 2006; Metselaar and de Jong van Lier, 2007; de Jong van Lier et al., 2009a; 2009b).

An empirical transpiration reduction function was proposed by Feddes et al. (1978) and is the reduction function most frequently used in combination with the SWAP simulation model (Kroes et al., 2008). One of its advantages is the computational simplicity, assuming water uptake to be partitioned over depth in proportion to root length density and applying an uptake reduction according to the soil water pressure head in respective soil layers and predefined threshold soil water pressure heads. These threshold values are hardly ever experimentally determined and usually taken from a compilation of values for several crops made by Taylor and Ashcroft (1972). It is, however, intuitive that not only crop type is determining for pressure head threshold values, but that parameters related to root system development, soil hydraulic properties, and meteorological conditions interact strongly with the involved processes. To understand these interactions, mechanistic modeling of these processes is imperative.

The spatial distribution of root length and of soil water status affects how much water is extracted from the soil by a root system, as well as how this uptake is partitioned over depth (Lascano and Van Bavel, 1984; Novák, 1987; Feddes and Raats, 2004). Under wet conditions, when soil hydraulic conductivity is high, the conductance within the plant

determines uptake rates (Passioura, 1988; Hopmans and Bristow, 2002), whereas root water uptake tends to be proportional to root length density (Novák, 1987). When one or more soil layers become drier and the hydraulic conductance of the soil-to-root pathway becomes the limiting factor to water uptake, root water uptake distribution will differ from the root length density distribution (Sharp and Davies, 1985).

Models for root water uptake at the single root scale, so-called microscopic or mesoscopic models, have been proposed by several authors since the 1950s (Philip, 1957; Gardner, 1960; Cowan, 1965; De Willigen and van Noordwijk, 1987; Raats, 2007). These models describe water uptake in terms of detailed calculation descriptions of water fluxes at and near the absorbing root surface. At the root system scale, macroscopic models (e.g., Hillel et al., 1976; Skaggs et al., 2006; Vogel et al., 2013) describe water uptake as a sink term in Richards' equation, assuming a continuum of soil, water, air, and roots. Both mesoscopic and macroscopic models can also provide insights into process sensitivity of system parameters including soil water retention and conductivity, root length density, and distribution. In general, macroscopic models require less detailed information and are more easily applied, whereas mesoscopic models allow prediction of the effects of small-scale phenomena, such as the reduction in hydraulic conductivity from bulk soil to root surface (Schröder et al., 2009) and specific aspects related to root system geometry (Doussan et al., 1998), but cannot easily be upscaled to the relevant scale, an entire plant, vegetation, or crop (Molz and Remson, 1971; Molz, 1981). This upscaling could involve a 3D root architecture model as developed by Diggle (1988), Pages and Aries (1988), or Lynch et al. (1997).

de Jong van Lier et al. (2008) attempted to upscale a single root (mesoscopic) model to the macroscopic level. Their model, based mainly on soil hydraulic properties, includes root length density as well as a composite of soil physical characteristics, the matric flux potential of the respective soil layers. The resulting reduction function showed consistent results when compared to some long-term experiments (de Jong van Lier et al., 2008). Comparisons to other experimental results (Casaroli et al., 2010; Faria et al., 2010) showed an overestimation of the predicted capacity of plants to extract soil water, when using the commonly accepted minimum root surface pressure head of -150 m, as simulations predict water uptake to occur at potential rates with much lower soil water pressure heads than experimentally observed.

In root water uptake models based on the soil-to-root pathway alone, including the one presented by de Jong van Lier et al. (2008), the hydraulic resistance within the plant is not considered. These models base their estimates solely on soil hydraulic resistance in combination with a limiting root water pressure head. This approach holds true under dry conditions, but is expected to fail when one or more soil layers are wet and internal plant resistances to water flow become limiting (Taylor and Klepper, 1975; Belmans

et al., 1979; Blizzard and Boyer, 1980). Within the plant, the radial flow in roots from the root–soil contact to the xylem vessels is considered to represent the greatest resistance (Frensch and Hsiao, 1993; Steudle and Peterson, 1998; Timlin et al., 2008). Among the water potentials within the plant, leaf water potential has been most frequently measured and used to assess plant water status (Turner, 1974; Brown et al., 1976; O'Toole and Cruz, 1980; Gupta et al., 1989; Cohen et al., 2005; Timlin et al., 2008; Suresh et al., 2010), partly because much of the early research in plant water relations was concerned with leaf processes (Jones, 1990). Leaf water potential is physically and physiologically related to root water potential (Jones, 1990).

Knowing the pressure head of the water inside the leaf, in the root xylem and at the root surface, the pressure head can be calculated if hydraulic resistances (radial and root-to-leaf longitudinal) and transpiration rates are known. This calculation is not straightforward because transpiration is one of the unknowns whose value is a supposed modeling outcome.

To overcome the problem, Heinen (2001) employed a relation proposed by Fisher et al. (1981), implying the introduction of two empirical parameters. However, in this study, we present an algorithm to estimate root water uptake, and actual and relative transpiration, without the use of an empirical relation such as Fisher et al. (1981), and without the introduction of new empirical parameters. The model will be tested against field data and model predictions will be compared to two other models.

Material and Methods

Theory

Considering root water uptake at the mesoscopic scale, we study a single root of radius r_0 extracting water from a soil cylinder (rhizosphere) with radius r_m . The xylem has a cylindrical shape with radius r_x and is centralized within the root.

The root tissue ($r_x \leq r \leq r_0$) is considered to have a constant water content θ_{root} ($\text{m}^3 \text{m}^{-3}$). The water content of the soil surrounding the root ($r_0 \leq r \leq r_m$) increases with distance from the root. At the root surface ($r = r_0$), a discontinuity in water content occurs (Fig. 1). Figure 1 also shows a typical shape of pressure head h vs. distance from the axial center. Unlike water content, pressure head is a continuous function of r . The derivative of h with r , however, is discontinuous at $r = r_0$ due to the different hydraulic properties of soil and root.

Matric flux potential M ($\text{m}^2 \text{d}^{-1}$) is defined as a function of unsaturated hydraulic conductivity $K(h)$ (m d^{-1}) and pressure head h (m) as:

$$M = \int_{h_w}^b K(h) dh \quad [1]$$

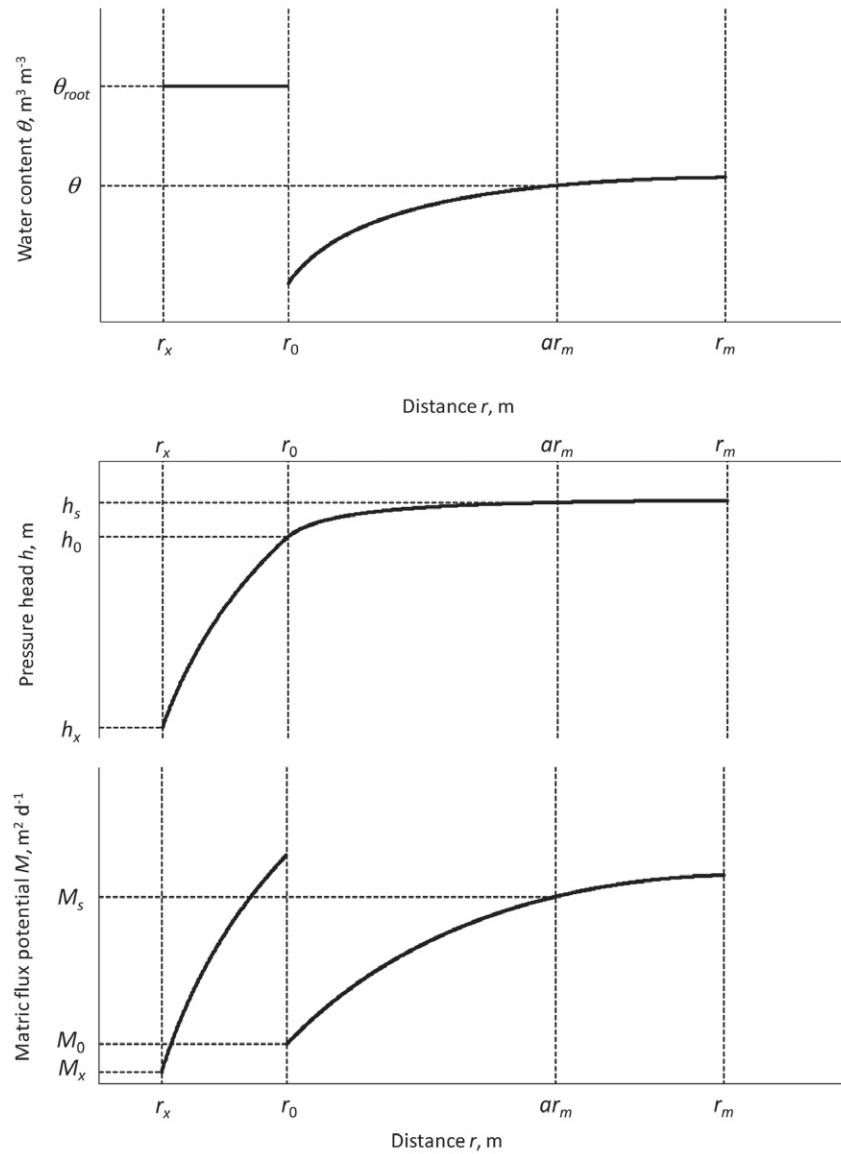


Fig. 1. Typical shapes of water content θ , pressure head h and matric flux potential M as a function of distance from axial center r in the rhizosphere and within the root. Subscripts x, 0, s, and m denote the outer xylem border, root surface, average soil, and rhizosphere extension, respectively. ar_m is the distance where average soil water conditions occur.

with h_w (m) being the pressure head at permanent wilting. Within the root we assume water content to be constant, hence $K(h) = K_{\text{root}}$ does not vary with h . Therefore, within the root:

$$M = \int_{h_w}^b K(h) db = K_{\text{root}} \int_{h_w}^b db = K_{\text{root}} (b - h_w) \quad [2]$$

For horizontal flow, the Darcy equation can be written in terms of M as:

$$q = -\frac{dM}{dr} \quad [3]$$

Analogous to the solution of the Richards equation for a layered soil presented by Ross and Bristow (1990), we use the fact that the first derivative of M as a function of r is a continuous function, whereas the function itself is discontinuous at the transition from soil to root (at $r = r_0$). An illustration of an M - r relation around a water extracting root is also shown in Fig. 1.

Assuming no water content changes within the root tissue, from the continuity equation for one-dimensional axisymmetric flow, it follows that for $r_x \leq r \leq r_0$:

$$\frac{\partial \theta}{\partial t} = -\frac{q}{r} - \frac{\partial q}{\partial r} = 0 \quad [4]$$

Combining Eq. [3] and [4] results in:

$$\frac{\partial M}{r \partial r} + \frac{\partial^2 M}{\partial r^2} = 0 \quad [5]$$

for which the following general solution is found:

$$M = C_1 \ln r + C_2 \quad [6]$$

where C_1 and C_2 are integration constants. The integration constants are defined by the following boundary conditions:

$$M = M_x; r = r_x \quad [7]$$

$$\frac{dM}{dr} = S_i \frac{r_m^2}{2r_0}; r = r_0 \quad [8]$$

where M_x ($\text{m}^2 \text{d}^{-1}$) is the matric flux potential in the xylem, r_x , r_0 , and r_m (m) are the xylem, root, and rhizosphere radii, respectively, and S_i ($\text{m}^3 \text{m}^{-3} \text{d}^{-1}$) is the root water uptake per unit of volume for the respective soil layer i . Equation [8] implies upscaling from the single root scale to the soil layer scale considering a homogeneous root distribution, i.e., each single root explores an equal soil volume. The relation between r_m and root length density R_L (m m^{-3}) is then given by:

$$r_m = \sqrt{\frac{1}{\pi R_L}} \quad [9]$$

To upscale from the soil layer to the root system scale, values of S_i for all Z layers are summed up yielding the total uptake which equals T_a :

$$T_a = \sum_{i=1}^Z S_i z_i \quad [10]$$

in which z_i (m) is the thickness of layer i .

Equation [8], substituting S_z by T_p/z , has been used by de Jong van Lier et al. (2006) and we refer to their publication for more details. Applying the boundary conditions from Eq. [7] and [8] results in:

$$C_1 = \frac{S_z}{2} r_m^2 \quad [11]$$

$$C_2 = M_x - \frac{S_z}{2} r_m^2 \ln r_x \quad [12]$$

Combining Eq. [6], [11], and [12] yields the following expression for M as a function of r ($r_x \leq r \leq r_0$):

$$M = M_x + \frac{S_z}{2} r_m^2 \ln \frac{r}{r_x} \quad [13]$$

and, combining to Eq. [2] we find, for $r_x \leq r \leq r_0$:

$$b = b_x + \frac{S_z r_m^2 \ln \frac{r}{r_x}}{2K_{\text{root}}} \quad [14]$$

Applying Eq. [14] at $r = r_0$ the following expression is obtained for b_0 :

$$b_0 = b_x + \frac{S_z r_m^2 \ln \frac{r_0}{r_x}}{2K_{\text{root}}} \quad [15]$$

Note that b is continuous over r (Fig. 1), therefore b_0 from Eq. [15] represents b at the exact interface between root and soil, irrespective of whether we consider this interface to assume the hydraulic properties of the root or soil.

Xylem water pressure head relates to leaf water pressure head b_ℓ (m) according to:

$$b_x = b_\ell + \frac{T_a}{L_1} \quad [16]$$

in which L_1 (d^{-1}) is the conductance over the root-to-leaf pathway. Combining Eq. [15] and [16] yields:

$$b_0 = b_\ell + \frac{T_a}{L_1} + \frac{S_z r_m^2 \ln \frac{r_0}{r_x}}{2K_{\text{root}}} \quad [17]$$

For the soil-to-root pathway, de Jong van Lier et al. (2008) developed the following relation between the bulk soil matric flux potential M_s ($\text{m}^2 \text{d}^{-1}$) and S_z :

$$S_z = \frac{4(M_s - M_0)}{r_0^2 - a^2 r_m^2 + 2(r_m^2 + r_0^2) \ln \frac{a r_m}{r_0}} = \rho(M_s - M_0) \quad [18]$$

where M_0 ($\text{m}^2 \text{d}^{-1}$) is the soil matric flux potential at the root surface.

The a parameter from Eq. [18] is defined by de Jong van Lier et al. (2008) and in this paper we use their value $a = 0.53$; thus ρ (m^{-2}) is given by:

$$\rho = \frac{4}{r_0^2 - a^2 r_m^2 + 2(r_m^2 + r_0^2) \ln \frac{ar_m}{r_0}} \quad [19]$$

Substituting Eq. [18] into Eq. [17] yields:

$$h_0 = h_\ell + \varphi(M_s - M_0) + \frac{T_a}{L_1} \Rightarrow h_0 + \varphi M_0 = h_\ell + \varphi M_s + \frac{T_a}{L_1} \quad [20]$$

$$\Rightarrow T_a = L_1 [h_0 - h_\ell + \varphi(M_0 - M_s)]$$

with φ (d m^{-1}) defined by:

$$\varphi = \frac{\rho r_m^2 \ln \frac{r_0}{r_x}}{2K_{\text{root}}} \quad [21]$$

The above approach assumes the soil hydraulic properties to be constant over the entire pathway through soil to root, and does not account for irregularities caused by an imperfect soil–root contact (Passioura, 1988; Carminati et al., 2009) or modified soil hydraulic properties in the near vicinity of roots (Carminati et al., 2010; Moradi et al., 2011). It also presupposes that macroscopically evaluated soil hydraulic properties can be used to represent the scale at which transport of water to roots is modeled, as discussed in detail by Metselaar and de Jong van Lier (2011).

An alternative approach for predicting root water uptake including soil and root resistance was presented by Couvreur et al. (2012). Comparing our Eq. [20] to Couvreur et al.'s Eq. [12] shows T_a to be a function of the difference between leaf and soil–root interface pressure heads multiplied by the root conductance in both models. The difference between both models resides in the way the soil-to-root resistance is included. In our model (Eq. [20]), a term with matric flux potentials is present to account for this resistance, whereas Couvreur et al. (2012) do not include it analytically but by solving the Richards equation numerically.

Parameterizing root conductivity and root conductance

The physical quantity *conductivity* (K_{root} , m d^{-1}) from the previous section, was used to express the water conductive property of root tissue for radial movement from the outer radius r_0 to the xylem radius r_x , whereas for the root-to-leaf pathway, the overall axial conductance L_1 was used. Reported values for both the radial and the axial parts of the pathway normally refer to the overall hydraulic *conductance* L (d^{-1}), or sometimes to the conductance multiplied by the root area per unit length of root L' (m d^{-1}). In

our approach, however, we express the matric flux potential as a function of r , and therefore we need K_{root} to express h , as a function of r along the radial pathway (Eq. [14]).

To establish the relation between K_{root} and L , we first compute root surface flux density q_0 using conductance:

$$q_0 = L(h_0 - h_x) \Leftrightarrow L = \frac{q_0}{(h_0 - h_x)} = \frac{K_{\text{root}} \frac{dh}{dr} \Big|_0}{(h_0 - h_x)} \quad [22]$$

where $dh/dr|_0$ is the pressure head gradient at the soil–root interface (at radial distance r_0).

By definition:

$$h_0 - h_x = \int_x^0 \frac{dh}{dr} dr \quad [23]$$

Considering mass conservation in the axisymmetric geometry within the root at constant K_{root} ,

$$q = q_0 \frac{r_0}{r} \Rightarrow \frac{dh}{dr} = \frac{dh}{dr} \Big|_0 \frac{r_0}{r} \quad [24]$$

and combining Eq. [23] and [24] results in:

$$h_0 - h_x = \frac{dh}{dr} \Big|_0 r_0 \int_x^0 \frac{dr}{r} = \frac{dh}{dr} \Big|_0 r_0 \ln \left(\frac{r_0}{r_x} \right) \quad [25]$$

Substitution of Eq. [25] into Eq. [22] results in the following relation between L and K_{root} :

$$K_{\text{root}} = L r_0 \ln \left(\frac{r_0}{r_x} \right) \quad [26]$$

Radial root conductance L has been measured by Rowse and Goodman (1981) for broad bean (*Vicia faba* L.); Sands et al. (1982) for common bean (*Phaseolus vulgaris* L.); Frensch and Steudle (1989) for maize (*Zea mays* L.); North and Nobel (1996) for the cactus species *Opuntia ficus-indica*; and Barrowclough et al. (2000) for onion (*Allium cepa* L.). These studies focused on observing variations in L as a result of root age, position on the root, or water stress history, and the reported L values are mostly in the order of 1 to $3 \cdot 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$; equivalent to about 0.8 to $2.5 \cdot 10^{-4} \text{ d}^{-1}$.

Reported root diameters of annual crops are mostly in the order of 0.1 to 1 mm (Gahoonia et al., 1997; Costa et al., 2002; Himmelbauer et al., 2004) and in our analyses, we used $r_0 = 0.5$ mm. Xylem diameter is not commonly reported in literature, but from root morphology images, it can be seen that r_x/r_0 is normally within the range of 0.1 to 0.9 making $\ln(r_0/r_x)$ between 0.1 and 2.3. Applying Eq. [26], we now find values of K_{root} to be in the range from $4 \cdot 10^{-10} \text{ m d}^{-1}$ to $3 \cdot 10^{-7} \text{ m d}^{-1}$. In our simulations, we used $K_{\text{root}} = 3.5 \cdot 10^{-8} \text{ m d}^{-1}$.

With respect to L_1 , the conductance over the root-to-leaf pathway, data from Zhuang et al. (2001) obtained for maize plants suggest a positive correlation between L_1 and T_p (De Willigen et al., 2011), resulting in values ranging from $0.7 \cdot 10^{-4} \text{ d}^{-1}$ for $T_p = 2 \text{ mm d}^{-1}$ to $2.1 \cdot 10^{-4} \text{ d}^{-1}$ for $T_p = 10 \text{ mm d}^{-1}$; in our simulations we used $L_1 = 1 \cdot 10^{-4} \text{ d}^{-1}$.

Analytical Application for a Special Case of Brooks and Corey Soils

The application of Eq. [20] requires a formulation of the $M(h)$ function to solve for h_0 and M_0 . In soils with soil hydraulic functions as defined by the Brooks and Corey (1964) equation, effective saturation Θ is described by:

$$\Theta = \left(\frac{h_b}{h} \right)^\lambda \quad [27]$$

in which h_b (m) is the air-entry pressure head and λ is a shape parameter ($\lambda > 0$). Applying the predictive approach by Burdine (1953) to the Brooks and Corey (1964) equation yields the following expression for $K(h)$:

$$K(h) = K_s \left(\frac{h_b}{h} \right)^{2+3\lambda} \quad [28]$$

Integration of K with respect to h to yield M (Eq. [1]) gives:

$$M(h) = \Lambda \left[\left(\frac{h_b}{h} \right)^p - \left(\frac{h_b}{h_w} \right)^p \right] \quad [29]$$

with p and Λ ($\text{m}^2 \text{ d}^{-1}$) defined by:

$$p = 1 + 3\lambda \quad [30]$$

$$\Lambda = -\frac{K_s h_b}{p} \quad [31]$$

Combining Eq. [20] and [29] results in:

$$h_0 + \varphi \Lambda \left(\frac{h_b}{h_0} \right)^p = h_\ell + \varphi M_s + \varphi \Lambda \left(\frac{h_b}{h_w} \right)^p + \frac{T_a}{L_1} \quad [32]$$

This equation is of the type:

$$\frac{A}{|h_0|^p} - |h_0| = B \quad [33]$$

with A (m^{1+p}) and B (m) defined by:

$$A = \varphi \Lambda |h_b|^p \quad [34]$$

$$B = h_\ell + \varphi M_s + \frac{A}{|h_w|^p} + \frac{T_a}{L_1} \quad [35]$$

No general solution for h_0 from Eq. [33] exists, except for the special case of an integer p . Relevant cases are $p = 2, 3, 4, \dots$. For $p = 2$ ($\lambda = 1/3$) a relatively simple solution is available and this case was further analyzed. For this case:

$$h_0 = \frac{B - \frac{P}{\sqrt[3]{2}} - \frac{\sqrt[3]{2} B^2}{P}}{3} \quad [36]$$

with P (m) defined by:

$$P = \sqrt[3]{3\sqrt{3}\sqrt{27A^2 - 4AB^3} + 27A - 2B^3} \quad [37]$$

Equation [36] and [37] yield real solutions for any $27A^2 - 4AB^3 \geq 0$. Combinations of M_s and h_x that, under a given root geometry (defined by r_x, r_0 , and r_m), and soil and root hydraulic properties that do not meet this condition, cannot be evaluated by these equations.

Numerical Implementation for van Genuchten Soils in the SWAP model

To evaluate the predictions of root water uptake and actual transpiration of the presented theory in multilayer scenarios together with other soil water balance components and different soil hydraulic functions, Eq. [20] was introduced into the ecohydrological model SWAP (Kroes et al., 2008) as a new subroutine. The SWAP method is designed to use the van Genuchten (1980) hydraulic functions for $\Theta(h)$ and $K(\Theta)$:

$$\Theta = \left[1 + (\alpha h)^n \right]^{-m} \quad [38]$$

$$K = K_s \Theta^l \left[1 - \left(1 - \Theta^{\frac{n}{n-1}} \right)^{1 - \frac{1}{n}} \right]^2 \quad [39]$$

in which h is the absolute value of the pressure head, $\Theta = (\theta - \theta_r)/(\theta_s - \theta_r)$ is the effective saturation, θ , θ_r and θ_s are water content, residual water content and saturated water content, respectively, and α (m^{-1}), n , and l are fitting parameters.

To allow variation of pressure head, soil hydraulic functions, and root length density with depth, Eq. [18] is spatially discretized referring to a spatial compartment i :

$$S_i = \rho_i (M_{s,i} - M_{0,i}) \quad [40]$$

with ρ_i (m^{-2}) defined by:

$$\rho_i = \frac{4}{r_0^2 - a^2 r_{m,i}^2 + 2(r_{m,i}^2 + r_0^2) \ln \frac{ar_{m,i}}{r_0}} \quad [41]$$

In this way, the root length density is defined per compartment, but for simplicity the root diameter r_0 is assumed to have one single value—model sensitivity to r_0 is low in similar models (de Jong van Lier et al., 2008)—which will be discussed later with respect to the present model. The matric flux potential at the root surface is calculated from indexed Eq. [20]:

$$h_{0,i} + \varphi_i M_{0,i} = h_\ell + \varphi_i M_{s,i} + \frac{T_a}{L_1} \quad [42]$$

with φ_i (d m^{-1}) defined analogous to Eq. [21] assuming constant values for xylem radius r_x and root conductivity K_{root} :

$$\phi_i = \frac{\rho_i r_{m,i}^2 \ln \frac{r_0}{r_x}}{2K_{\text{root}}} \quad [43]$$

Note that both $h_{0,i}$ and $M_{0,i}$ vary with depth, as they depend on the depth-specific $M_{s,i}$, $\varphi_{s,i}$ (defined by depth-specific root density) and S_i . The pressure head in the xylem h_x is calculated by Eq. [16] (incorporated in Eq. [20] and [42]) and is assumed to be constant in the root zone. A similar assumption was made and justified by de Jong van Lier et al. (2008) and implies neglecting within-root pressure head differences due to gravity or axial root resistances.

Equation [42] contains the unknowns T_a , h_ℓ , and $h_{0,i}$, whereas $M_{0,i}$ is a function of $h_{0,i}$. For van Genuchten (1980) type soils, no straightforward expression for $M(h)$ exists, but a converging series approximation is available (de Jong van Lier et al., 2009a).

Substitution of $M_{0,i}$ in Eq. [40] gives the root water uptake S_i at a particular node. The actual transpiration follows from the integration of uptake rates (Eq. [10]).

To solve Eq. [42], the correct combination of h_ℓ and T_a for which $h_{0,i}$ yields T_a must be found. Furthermore, T_a is a function of h_ℓ , less negative values of h_ℓ result in a lower outcome for T_a at a given distribution of M_s over depth. The new SWAP subroutine imposes a limiting (or minimum) leaf water pressure head h_w . Then $T_r(h_\ell)$ is defined by a step function according to:

$$\begin{aligned} T_r &= 1 & h_\ell > h_w \\ 0 \leq T_r &\leq 1 & h_\ell = h_w \\ T_r &= 0 & h_\ell < h_w \end{aligned} \quad [44]$$

This procedure is similar to the approach used by Couvreur et al. (2012) in establishing a threshold value for the collar pressure head. It follows from Eq. [44] that T_r is undefined for $h_\ell = h_w$, and as both the leaf water pressure head h_ℓ and the actual transpiration T_a from Eq. [42] are unknown, the equation system has to be solved iteratively. To optimize the numerical procedure, at first the occurrence of water stress ($T_a < T_p$) or not ($T_a = T_p$) is determined. To do so, the new SWAP subroutine calculates a maximum achievable total uptake (T_m , m d^{-1}), corresponding to $h_\ell = h_w$.

If $T_m \geq T_p$ actual transpiration is calculated using the boundary conditions:

$$T_a = T_p \quad ; \quad h_\ell > h_w \quad [45]$$

Assuming a value for h_ℓ , the corresponding T_a is calculated using Eq. [40]–[43] and [10]. If T_a turns out to be smaller than T_p , the value of h_ℓ is decreased; if $T_a > T_p$, the value of h_ℓ is increased. Employing the Newton–Raphson iterative procedure, the value of h_ℓ can be found in a computationally very efficient way.

On the other hand, if $T_m < T_p$, the actual transpiration is calculated with the constraint $h_\ell = h_w$. We then estimate an actual transpiration T_a^* based on the previous conditions and calculate the resulting T_a with Eq. [40]–[43]. If $T_a < T_a^*$ the next step estimate T_a^* is made smaller; if $T_a > T_a^*$, the value of T_a^* is increased for the next iteration. This is also an iterative procedure which (in the current implementation) is solved using the Newton–Raphson algorithm.

Although the overall procedure was implemented efficiently as a SWAP subroutine, the fact that Eq. [44] does not attribute a unique value to T_r for $h_\ell = h_w$ may impose some numerical restrictions. If L_1 or K_{root} are specified out of their expected physical range (either very high or very low values), the Newton–Raphson iterative procedure may fail to converge. Using values of L_1 and K_{root} within the normal range convergence is normally established

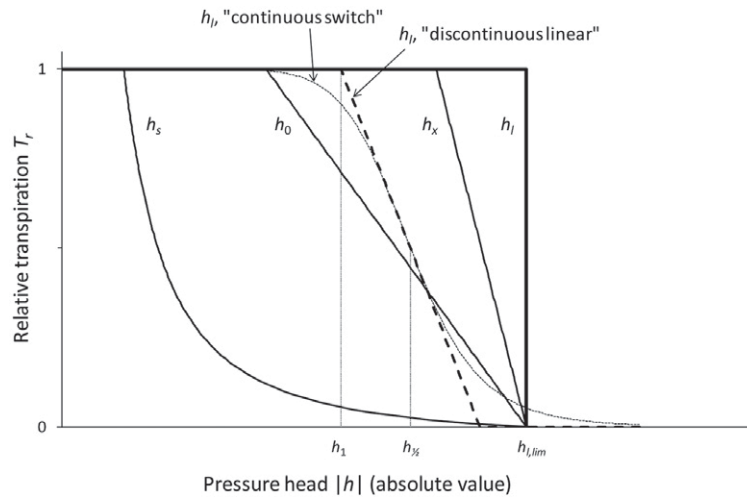


Fig. 2. Schematic representation of relative transpiration as a function of pressure head in the leaf (h_l), in the xylem (h_x), at the root surface (h_0), and average soil pressure head (h_s), considering presented theory. Also shown is a typical shape of leaf pressure head h_l according to the discontinuous linear and continuous switch models presented by Fisher et al. (1981) (Eq. [46] and [47]).

in a few iterations. As an alternative, Fisher et al. (1981) present a “discontinuous linear model” according to which:

$$T_r = \begin{cases} 1 & h_l > h_1 \\ \frac{h_l + h_1 - 2h_{1/2}}{2h_1 - 2h_{1/2}} & 2h_{1/2} - h_1 \leq h_l \leq h_1 \\ 0 & h_l < 2h_{1/2} - h_1 \end{cases} \quad [46]$$

with $h_{1/2}$ (m) the leaf water pressure head at which $T_r = 0.5$ and h_1 (m) the highest (less negative) leaf water pressure head for which $T_r = 1$. They also present a “continuous switch model,” used by Campbell (1991) and Kremer et al. (2008):

$$T_r = \frac{1}{1 + \left(\frac{h_l}{h_{1/2}} \right)^a} \quad [47]$$

Unlike Eq. [44], both equations by Fisher et al. (1981) define T_r as a function of h_l over the entire range. Fisher et al. (1981) report experimental values of $h_{1/2}$ around -70 m, with a in the order of 5; Campbell (1991) suggests 10 as a typical value for a . Values for h_w , measured as limiting leaf water pressure head, were reported by Turner (1974) around -170 m for maize, -200 m for sorghum [*Sorghum bicolor* (L.) Moench] and -130 m for tobacco (*Nicotiana tabacum* L.), whereas O’Toole and Cruz (1980) found values between -80 and -120 for rice (*Oryza sativa* L.).

The discontinuous linear model (Eq. [46]) and the continuous switch model (Eq. [47]) by Fisher et al. (1981) result in a smoother transition of the constant rate phase to the falling rate phase than the Eq. [44] implemented in the new SWAP subroutine. Figure

2 gives the shape of T_r as a function of h_l according to the proposed model used in SWAP simulations (Eq. [44]) and Eq. [46] and [47] proposed by Fisher et al. (1981). Also shown are typical shapes of h_x , h_0 , and h_s , according to the presented theory in which h_x , h_0 , and h_l are linearly correlated and depend on uptake rate S or actual transpiration T_a (Eq. [15] and [16]), whereas h_s is defined as function of h_0 by the relation between the respective matrix flux potentials and S (Eq. [18]). Comparing the proposed reduction function implemented in our SWAP simulations to the relations presented by Fisher et al. (1981), Fig. 2 shows the similarity between h_l according to Fisher et al. (1981), especially their discontinuous linear function, and h_x or h_0 used in the SWAP simulations. Using Eq. [44], as in the simulations presented here, instead of Eq. [46] or [47] used by Heinen (2001), the number of empirical parameters is reduced by 1. These parameters are difficult to obtain and their effect on predictive quality is difficult to evaluate. Many other root water uptake models use a limiting value of pressure head in the root xylem or at the collar (Doussan et al., 1998; Schröder et al., 2009; Couvreur et al., 2012).

Comparison of SWAP Simulations to Experimentally Obtained Data

The adapted version of the SWAP ecohydrological model was used to simulate the soil water balance and relative transpiration for two field experiments performed in Brazil. The field experiments comprised an area of approximately 500 m^2 in the municipality of Piracicaba, São Paulo State, Brazil ($22^\circ 42' 30''$ S lat, $47^\circ 38' 0''$ E long, 546 m a.s.l.) The soil at the location is a Rhodic Kanhapludalf according to the USDA/Soil Survey Staff classification, or a Rhodix Lixisol according to the FAO WRB map of World Soil Resources. The soil has a bulk density of 1560 kg m^{-3} in the Ap horizon ($0\text{--}0.2 \text{ m}$) and 1380 kg m^{-3} in the Bt horizon ($0.2\text{--}0.8$

m). The texture is clayey, with clay contents in the two horizons of 0.58 and 0.68 kg kg⁻¹, respectively.

An automatic meteorological station of the University of São Paulo is located within 100 m of the experimental site and provided meteorological data during the experiment. Reference evapotranspiration (ET₀, mm d⁻¹) was calculated by the FAO Penman–Monteith method (Allen et al., 1998).

Between June and September 2010, under very dry conditions, common bean was grown. The plot was sprinkler irrigated during the initial growing stages (until 2 August) but subject to water stress after that, in the reproductive phase. During the 91 d of the field experiment (15 June–13 September), rainfall was observed on 13, 14, and 15 July (total height 62.9 mm) and after that only on 7 September (12.8 mm) during the ripening period. To guarantee the survival of the crop, one irrigation gift was applied after 2 August, on 24 August. Soil water content was monitored at three depths (5, 15, and 30 cm) every 30 min, using Decagon EC-5 sensors. The EC-5 sensor determines volumetric soil water content by measuring the dielectric constant of the media using capacitance/frequency domain technology at a frequency of 70 MHz. Sensors are 8.9 cm long (of which 5 cm is the active part in contact with soil). For mineral soils, accuracy is $\pm 0.03 \text{ m}^3 \text{ m}^{-3}$ with a resolution of $0.001 \text{ m}^3 \text{ m}^{-3}$. Canopy temperature was measured by an automated infrared thermometer (Apogee Instruments, model SI-111[®]) installed at 2 m height with a horizontal angle of 60° and a half angle of the field of view of 22°, resulting in a target observation area of 65 m². Sensor accuracy is $\pm 0.2^\circ\text{C}$ and measuring range is from -10°C to 65°C . Readings were performed every 30 min, both in the stressed plots and in a neighboring plot which was irrigated to maintain potential transpiration.

From November 2010 to March 2011, soybean [*Glycine max* (L.) Merr.] was grown on the same field; meteorological conditions were very different from the first period, with significant rainfall on more than 60% of the days. Total rainfall was 974 mm, almost twice the potential evapotranspiration for the same period. During this experiment, EC-5 sensors were available at 15- and 30-cm depth.

Simulations were performed using the SWAP model with three reduction functions: (i) the Feddes et al. (1978) reduction function (FRF) with threshold pressure heads according to values reported by Taylor and Ashcroft (1972) and listed in Table 3; (ii) the de Jong van Lier et al (2008) reduction function (JRF): with input values listed in Table 1; and (iii) the proposed reduction function (PRF): the reduction function proposed above which includes radial resistance, with input values listed in Table 1. Values for root length density R_L were based on field observations at three depths (5, 15, and 30 cm). The exponential equation used by Li et al. (2001) was fitted to these data for the Common Bean scenarios. Used values of R_L vs. depth are shown in Fig. 3.

Table 1. Parameter values used in examples with the Brooks and Corey (1964) soils special case (BC) and in Soil–Water–Atmosphere–Plant (SWAP) simulations with van Genuchten (1980) soils using the de Jong van Lier et al. (2008) reduction function (JRF) and proposed reduction function (PRF) (VG).

Parameter	Value	Unit	Case
b_b	−0.05 m	m	BC
λ	1/3		BC
p	2		BC
K_s	0.5	m d ^{−1}	BC
r_m	0.02	m	BC
R	0.08	cm cm ^{−3}	BC
r_x	0.2	mm	BC and VG(PRF)
r_0	0.5	mm	BC and VG(JRF and PRF)
K_{root}	3.5×10^{-8}	m d ^{−1}	BC and VG(PRF)
L_1	1×10^{-4}	d ^{−1}	BC and VG(PRF)
h_w	−200/−150	m	BC and VG(JRF and PRF)†

† −200 m for h_w (limiting leaf water pressure head) was used in BC and VG-PRF simulations; −150 m for h_w (limiting root surface pressure head) was used in VG-JRF simulations.

Soil water retention parameters θ_s , θ_r , α , and n were obtained by simultaneous observations of soil water content using the previously described EC-5 sensors and of pressure head using polymer tensiometers with a measuring range from 0 to −1600 kPa and an accuracy of 2.38 kPa (De Rooij et al., 2009). These measurement devices were installed in the experimental field and Eq. [38] was fitted to the obtained data.

Hydraulic conductivity parameters K_s and l were obtained by fitting data of an evaporation experiment (Wind, 1968) to Eq. [39]. The evaporation experiment was performed in a laboratory using polymer tensiometers, utilizing the methodology described in Durigon et al. (2011). Observations from both the field and evaporation experiment refer to the range of pressure heads between −150 m and −1 m. The soil hydraulic data obtained were used in the SWAP simulations and are shown in Table 2. In this context, θ_s and K_s should not be interpreted as representing saturated conditions, but merely as fitting parameters referring to values extrapolated from unsaturated experiments. Both parameters were used for the estimation of values under unsaturated conditions by Eq. [38] and [39], and the extrapolation may have biased simulation results during very wet periods ($h_s > -1$ m) which occurred during the soybean scenario.

The bottom boundary condition was unit hydraulic gradient. Soil evaporation was calculated by reduction to the maximum Darcy flux limited by the theory of Black cited in Kroes et al. (2008). Specific parameters for the Black evaporation formula were an evaporation coefficient of $0.35 \text{ cm d}^{-1/2}$ and a reset to potential rate after any precipitation $> 5 \text{ mm d}^{-1}$.

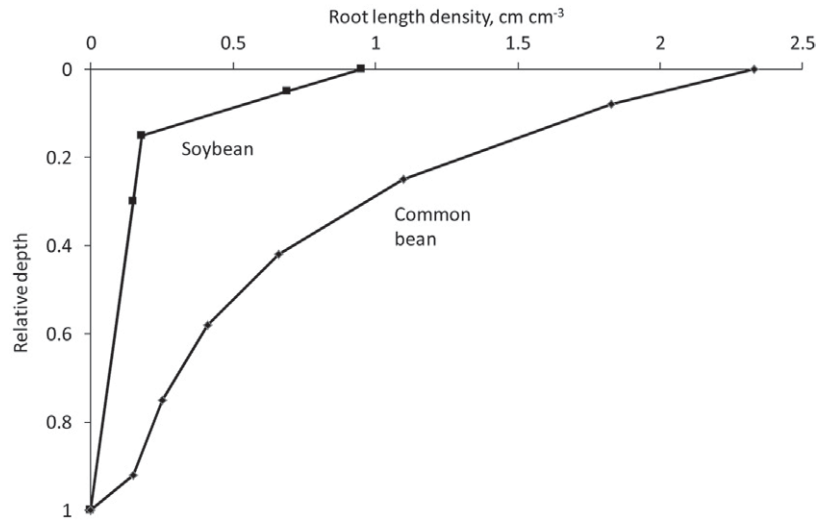


Fig. 3. Root length density distribution over depth for common bean (*Phaseolus vulgaris* L.) and soybean [*Glycine max* (L.) Merr.] used in simulations with the Soil–Water–Atmosphere–Plant (SWAP) model.

Table 2. Soil physical properties for the van Genuchten (1980) equations used in Soil–Water–Atmosphere–Plant (SWAP) simulations.

Depth	θ_r	θ_s^*	α	n	$K_s^\dagger$	l
cm			m^{-1}		$m\ d^{-1}$	
0–10	0	0.233	0.03305	1.2215	0.000145	5.2188
10–20	0	0.359	0.09050	1.0983	0.001556	1.9387
20–100	0.215	0.411	0.05737	1.1885	0.000440	−0.4750

† Values for θ_s and K_s refer to extrapolated values and are not meant to be interpreted as real values under saturated conditions; see text for details.

Results and Discussion

Brooks and Corey Soils Special Case Applied to a Monolayer Soil

Figure 4 shows $h(r)$ relations calculated for a homogeneously rooted soil profile and parameter values for the Brooks and Corey (BC) case from Table 1 by solving for h_0 using Eq. [36] and applying an iterative method to satisfy Eq. [20]. Figure 4 shows results for the bulk soil pressure head $h_s = -5$ m in combination with three leaf pressure heads, -20 , -50 , and -200 m. Calculated uptake rates are in the order of 1 to $2 \cdot 10^{-3}\ d^{-1}$, and pressure head values show soil resistance to be of most importance for the lowest (most negative) leaf water pressure head, whereas plant resistance becomes important when the leaf water pressure head is high.

For the same parameter values (Table 1), Fig. 5 shows the uptake rate, xylem pressure head and root surface pressure head as a function of leaf pressure head. The model shows how higher uptake rates in the wetter soil ($h_s = -5$ m) are driven by a larger difference between h_0 and h_x . When xylem pressure head approaches the

average soil pressure head, uptake drops to zero, and h_0 approaches h_x . In the wetter soil, the uptake curve shows an almost piecewise-linear behavior: an approximately linear increase in S between $h_\ell = h_s = -5$ m and $h_\ell \approx -40$ m, and an almost constant value of S when h_ℓ decreases to more negative values. In both cases, uptake rates tend to a finite maximum value $S_{\max}$ when leaf pressure head decreases to $-\infty$. This observation can be compared to the maximum transmission rate of water from a water table to the evaporation zone as analytically shown by Gardner (1958) and Ripple et al. (1972). In the case of a water extracting root in axisymmetric geometry:

$$q_r = q_0 \frac{r_0}{r} = -\frac{dM}{dr} \Leftrightarrow -dM = q_0 r_0 \frac{dr}{r} \quad [48]$$

Considering $M = M_0$ at $r = r_0$ and $M = M_s$ at $r = ar_m$, Eq. [48] can be integrated to yield:

$$M_s - M_0 = q_0 r_0 \ln \frac{r_0}{ar_m} \quad [49]$$

and in combination with Eq. [29], is valid for Brooks and Corey (1964) soils and gives:

$$\Lambda \left[\left(\frac{h_b}{h_s} \right)^p - \left(\frac{h_b}{h_0} \right)^p \right] = q_0 r_0 \ln \frac{r_0}{ar_m} \quad [50]$$

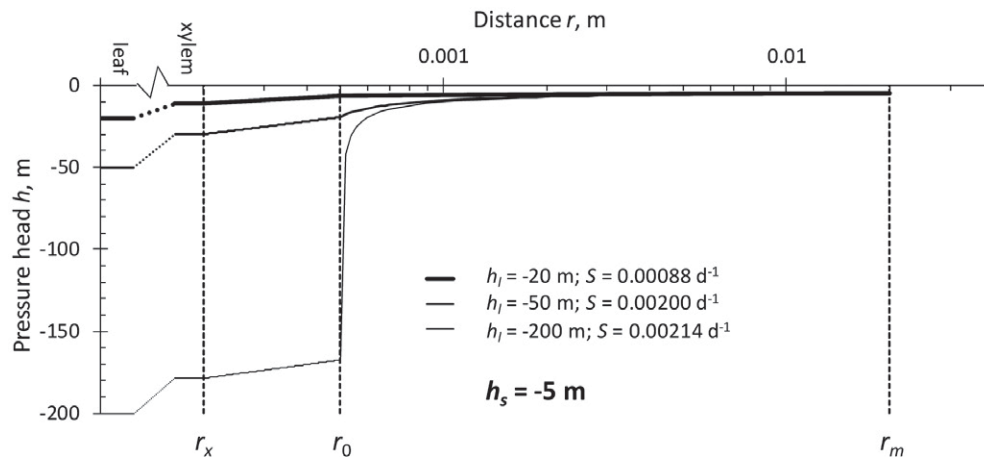


Fig. 4. Pressure head h as a function of distance from the axial center r around and within a water extracting root calculated for a Brooks and Corey (1964) soil by Eq. [36] with parameter values from Table 1. Results are shown for bulk soil pressure head $h_s = -5$ m in combination with three values of leaf pressure head h_l .

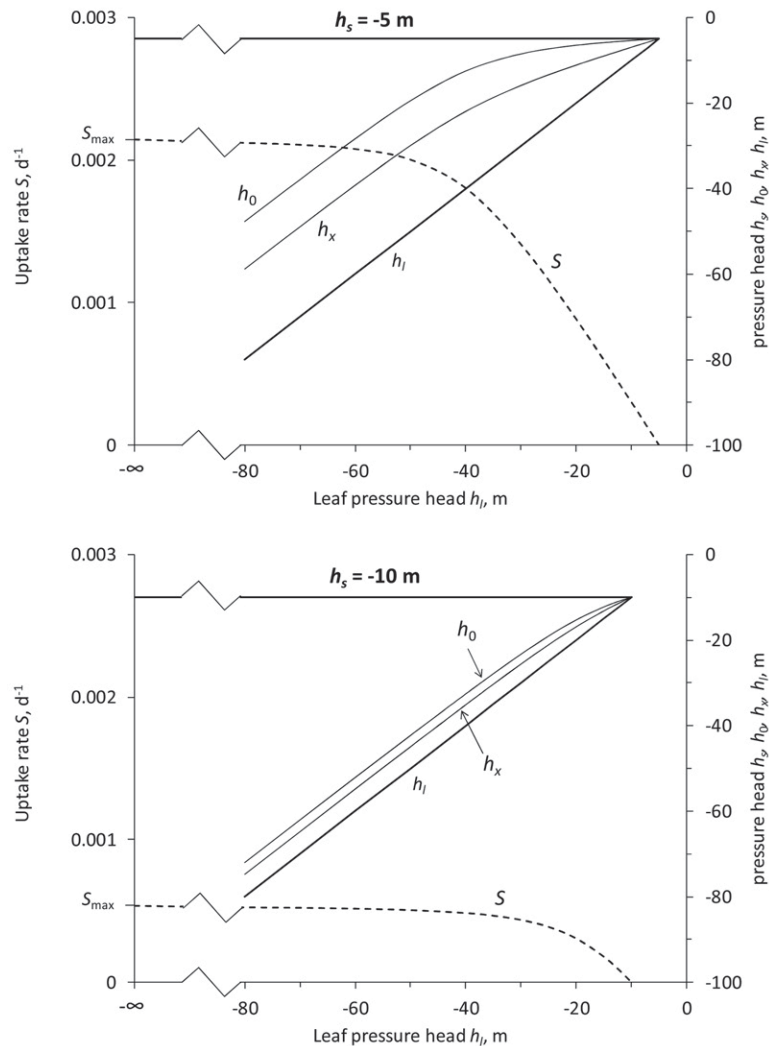


Fig. 5. Uptake rate S and pressure heads as a function of leaf pressure head h_l calculated for two average soil water pressure heads in a Brooks and Corey (1964) soil ($h_s = -5$ m and -10 m) by Eq. [36] with parameter values from Table 1.

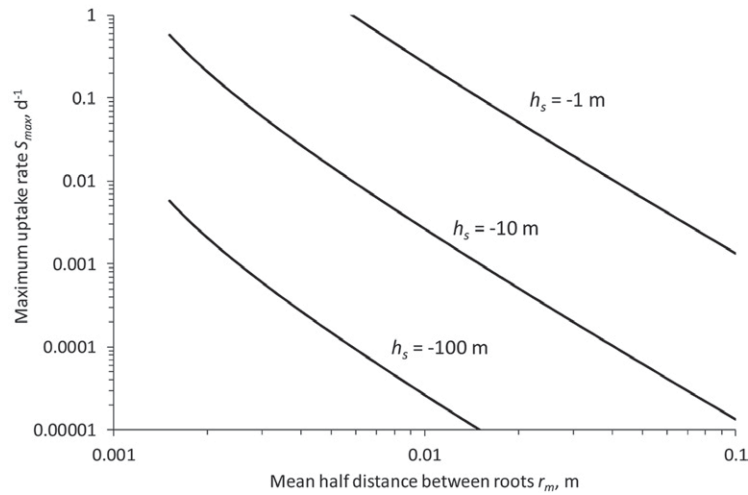


Fig. 6. Maximum uptake rate $S_{\max}$ as a function of mean half distance between roots r_m calculated for three average soil water pressure heads ($h_s = -1$ m, -10 m, and -100 m) in a Brooks and Corey (1964) soil by Eq. [53] with parameter values from Table 1.

Maximum uptake rates are reached when leaf, xylem, and root surface pressure heads h_ℓ , h_x , and h_0 tend to high negative values. Then, Eq. [50] reduces to:

$$\Lambda \left(\frac{h_b}{h_s} \right)^p = q_0 r_0 \ln \frac{r_0}{ar_m} \Leftrightarrow q_0 = \frac{\Lambda \left(\frac{h_b}{h_s} \right)^p}{r_0 \ln \frac{r_0}{ar_m}} \quad [51]$$

The relation between q_0 and S is given by:

$$S = -\frac{2q_0 r_0}{r_m^2} \quad [52]$$

and, in combination with Eq. [51] and [52], allows calculating the maximum uptake rate $S_{\max}$ (d^{-1}) in Brooks and Corey (1964) soils:

$$S_{\max} = \frac{2\Lambda \left(\frac{h_b}{h_s} \right)^p}{r_m^2 \ln \frac{ar_m}{r_0}} \quad [53]$$

Values of $S_{\max}$ as a function of r_m calculated by Eq. [53] for the same system parameters used in Fig. 4 and 5 are shown in Fig. 6 for three bulk soil water pressure head values h_s . Using $p = 2$, it follows from Eq. [53] that values of $S_{\max}$ increase by a factor 100 when h_s is divided by 10. In very dry soils ($h_s = -100$ m), considerable uptake rates can only be achieved by very dense root systems. Under such conditions, an uptake rate of $0.001 d^{-1}$ is possible with an r_m of about 0.003 m, corresponding to a root length density R_L of 3.5 cm cm^{-3} . In contrast, in a wet soil ($h_s = -1$ m), similar uptake rates can be achieved by a root system with $r_m = 0.1$ m, corresponding to $R_L = 0.003 \text{ cm cm}^{-3}$.

Root radius r_0 may show a relatively high variation within one root system. Therefore, to apply the presented theory, rough estimates must often be used and a low model sensitivity to root radius would be important. The sensitivity of the calculated uptake rate S to variations in root radius r_0 was evaluated over a large range (Fig. 7) for three values of soil water pressure heads ($h_s = -5$ m, -10 m, and -30 m). Xylem radius was assumed to be proportional to root radius by $r_x = 0.4r_0$, while the leaf water pressure head $h_\ell = -200$ m was used. Other parameter values were taken from Table 1. Figure 7 shows a low sensitivity of S to changes in r_0 . Over the full analyzed range, r_0 increases more than 100 times, whereas the corresponding increase in S is about three times for all three values of h_s evaluated.

Application of SWAP with van Genuchten Soils and Three Reduction Functions

Predicted water content using the SWAP model with FRF, JRF, and PRF, together with measured water content using the EC-5 sensors are shown in Fig. 8 for the three monitored depths of the common bean experiment and the two monitored depths of the soybean experiment. For the common bean experiment, predicted and observed values show good agreement, whereas predicted water content for the soybean experiment underestimate the observed values, especially at a depth of 0.30 m. Differences in predicted water content between the three reduction functions occur and reflect the different model approaches regarding transpiration reduction and water extraction over depth.

Simulated relative transpiration over the growing season obtained with the SWAP model in combination with the three reduction functions FRF, JRF, and PRF, are shown in Fig. 9 for the common bean scenario with standard input from Table 1 (Fig. 9a), perturbed input (Fig. 9b) and for the soybean scenario with standard

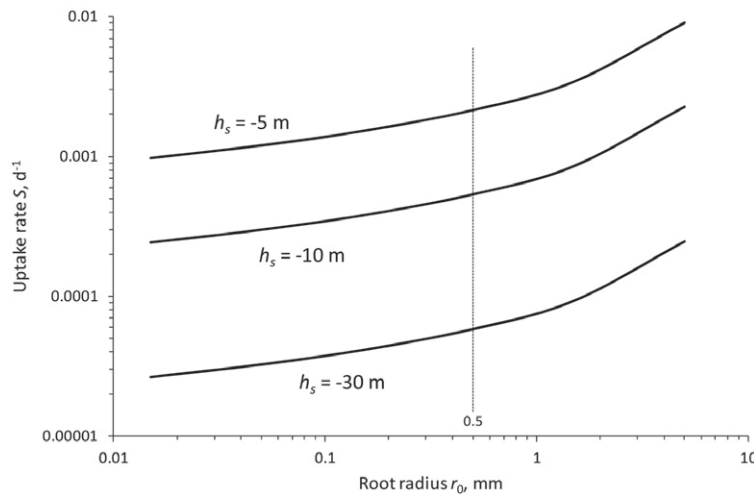


Fig. 7. Uptake rate S as a function of root radius r_0 calculated for three average soil water pressure heads ($h_s = -5$ m, -10 m, and -30 m) in a Brooks and Corey (1964) soil by Eq. [36] with $r_x = 0.4r_0$ and parameter values from Table 1.

input (Fig. 9c), together with rainfall and irrigation data. For the common bean scenario, open circles represent experimental T_r calculated from vapor pressure deficit and fully transpiring versus actual canopy temperature as determined between 5 Aug. and 1 Sept. 2010 using the Crop Water Stress Index (CWSI) proposed by Idso et al. (1981). In addition, T_r was considered equivalent to $(1 - \text{CWSI})$, as proposed by Jackson et al. (1981).

In the standard common bean scenario (Fig. 9a), with almost no rainfall and only a few small additional irrigations, all three models predict considerable transpiration reduction. Under dry circumstances, soil resistance can be expected to be higher than plant resistance and JRF and PRF do not differ greatly in relative transpiration predictions. When stress increases and T_r becomes very low, the PRF h_0 tends to values close to $h_\ell = h_w = -200$ m, whereas the most negative value in the JRF is $h_0 = h_w = -150$ m (see footnote in Table 1). The fact that the PRF allows root water pressure head to become more negative than the JRF explains why, during the very dry days, T_r is predicted to be somewhat higher using the PRF than using the JRF.

When irrigation is applied on a very dry soil, as on 24 August or in the very beginning of September, compensation is likely to occur. Compensation is the enhanced uptake of water from wetter layers when uptake from one or more dry layers is reduced (Li et al., 2001; Jarvis, 2011). Both JRF and PRF, implicitly computing compensation, predict an immediate recovery and return to potential transpiration rates ($T_r = 1$). The FRF, not including compensation, predicts continuing water stress, although alleviated, due to the fact that some of the (deeper) rooted soil layers remain dry. This same difference in modeling approach results in the prediction of water stress by the FRF for some days of the (wet) soybean scenario (Fig.

9c), where JRF and PRF predict no transpiration reduction at all during the entire soybean crop cycle.

For the common bean scenario, experimental values of T_r show a reduction to about 0.2 around 21 August, similar to the prediction of the three reduction functions. The onset of water stress, as well as the partial recuperation after the irrigation of 24 August appear to be better predicted by the FRF. At this point, it is important to note that predictions of all three reduction functions are a result of their respective algorithms together with the input values from Tables 1, 2, and 3. The quality of these models can be evaluated using several criteria, including comparing the goodness of fit after calibration, or by performing an independent subsystem experiment to determine and compare the calibrated parameter. This last approach can be used for parameters in mechanistic models like JRF or PRF, but is hardly applicable to empirical models like FRF.

To highlight the calibration flexibility of PRF, the effect of two perturbations of input parameters on PRF predictions was evaluated (Fig. 9b). In the first perturbation, h_w was increased from -200 m to -100 m. As a consequence, predicted transpiration reduction starts 5 d earlier and recovery after the irrigation of 24 August is not complete. Predicted T_r is closer to the experimental values, although the decline of relative transpiration over time around 10 August is much steeper than the experimental data. A second perturbation was performed by increasing plant hydraulic resistance 10 times (dividing K_{root} and L_1 from Table 1 by 10). This perturbation has an effect comparable to increasing h_w at the onset of transpiration reduction and also yields the prediction of incomplete recovery after an irrigation, but results in the prediction of a smaller reduction in T_r during the water stress period of the first 10 d, which agrees with the experimentally observed decline.

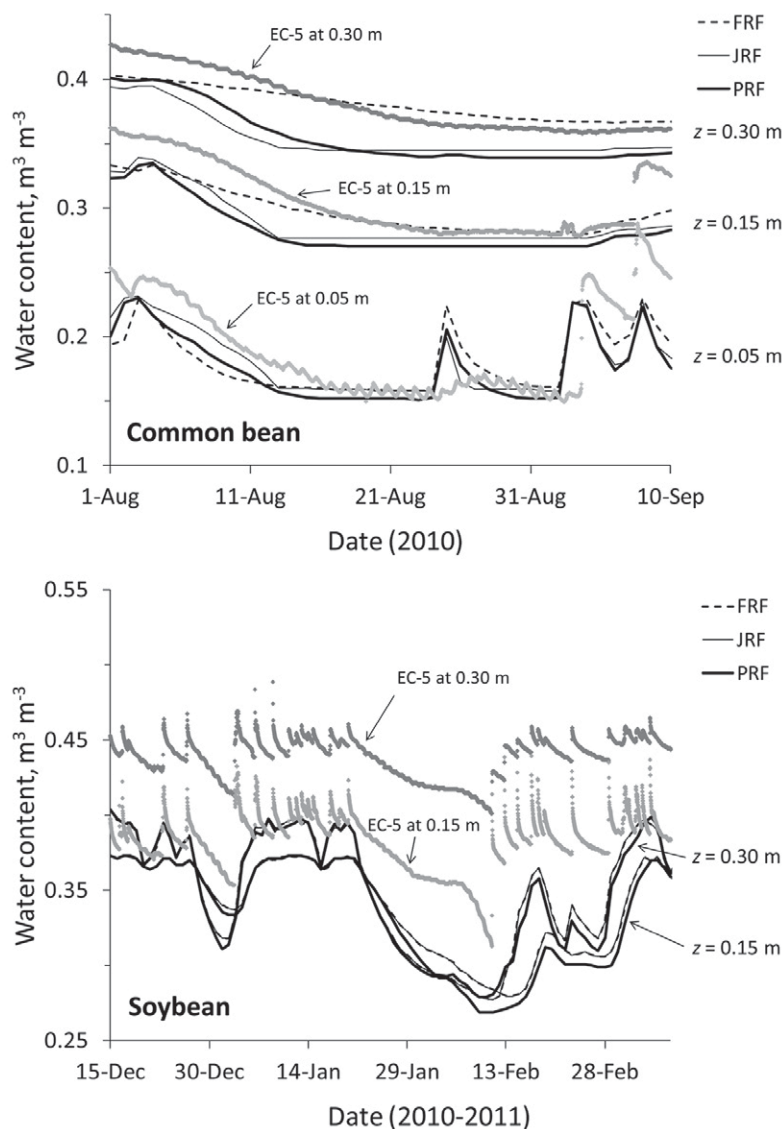


Fig. 8. Observed water contents using EC-5 sensors together with predicted water contents using the Soil–Water–Atmosphere–Plant (SWAP) model with Feddes et al. (1978) reduction function (FRF), de Jong van Lier et al. (2008) reduction function (JRF), and proposed reduction function (PRF) for (a) the common bean (*Phaseolus vulgaris* L.) (three depths) and (b) the soybean [*Glycine max* (L.) Merr.] (two depths) scenario.

It should be remembered that the original values and those used for perturbation are within the normal range and that the choice of parameter values allows the models to be fitted to experimental results. Comparing JRF to PRF, the former does not consider plant hydraulic resistance and tends to predict very abrupt changes in relative transpiration as noted by de Jong van Lier et al. (2008). The PRF allows smoother transitions, more in agreement with experimental values, as shown in Fig. 9b. The FRF is computationally much simpler and therefore preferred by many users as it provides good predictions, but its input parameters (threshold pressure heads and corresponding potential transpiration rates) are empirical and are expected to vary according to properties of the soil, plant, and atmosphere. Compilations of FRF parameter values (Taylor and Ashcroft, 1972) do not include aspects of root length density or

soil hydraulic properties. The PRF is an attempt to add detail to these relations and represents a challenge for future research into estimating FRF threshold values from PRF predictions.

Another aspect highlighting changes in the relative importance of soil and plant hydraulic resistance is illustrated in Fig. 10. It shows the sensitivity of model predictions of relative cumulative transpiration to interactions between root length density, radial root conductivity, and axial conductance. In the relatively high root length density scenario from Fig. 3, sensitivity to K_{root} and L_1 is lower than in a scenario with a 10 times sparser root system. For most of the scenarios, even with very high conductance or conductivity, the plant with a smaller root length density is not able to realize water uptake at the same rate as the densely rooted plant.

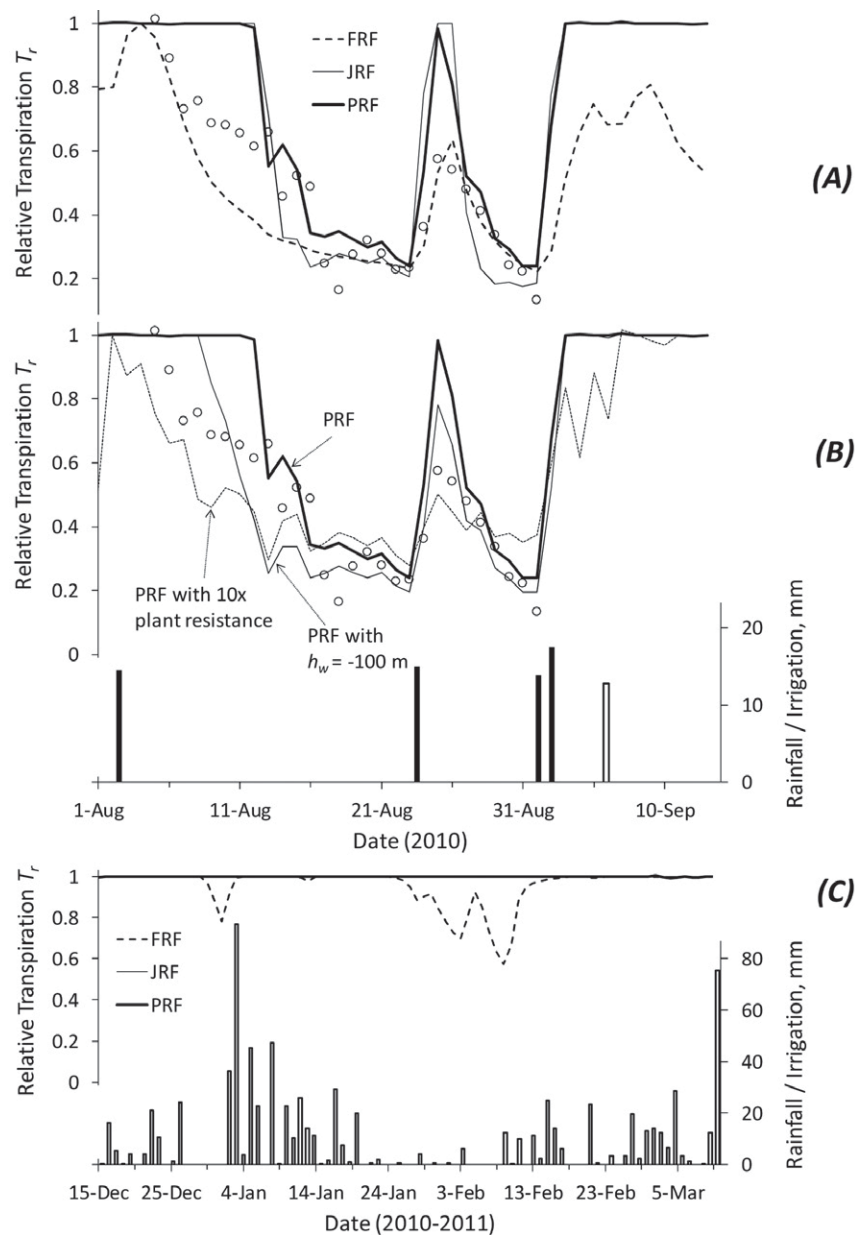


Fig. 9. Rainfall and irrigation and simulated relative transpiration over the growing season obtained with the Soil–Water–Atmosphere–Plant (SWAP) model for (a) the common bean (*Phaseolus vulgaris* L.) scenario with FRF, JRF, and proposed reduction function (PRF); (b) the common bean scenario with PRF using standard input as well as modifying h_w to -100 m or increasing plant resistance tenfold; and (c) the soybean [*Glycine max* (L.) Merr.] scenario with Feddes et al. (1978) reduction function (FRF), de Jong van Lier et al. (2008) reduction function (JRF), and proposed reduction function (PRF). Solid bars represent irrigation, open bars rainfall. Open circles represent experimentally determined T_r .

This is in agreement with the maximum uptake rates as shown by Eq. [53] and in Fig. 7.

Under dry circumstances, plant resistance, which is simulated by PRF but not in JRF, is negligible compared to soil resistance. As a result, both JRF and PRF predictions for T_r are very similar. The FRF predictions with no compensation mechanism are sometimes very different. However, even when total simulated uptake between the three model concepts is not dissimilar, as on 19 August during the common bean scenario, uptake distribution

Table 3. Threshold pressure heads for the Feddes reduction function (FRF) used in Soil–Water–Atmosphere–Plant (SWAP) simulations for the common bean and soybean scenarios.

	Common bean	Soybean
b_1	0	0
b_2	0	0
$b_{3,b}$ (corresponding to $T_p = 5 \text{ mm d}^{-1}$)	-7.5 m	-5 m
$b_{3,l}$ (corresponding to $T_p = 1 \text{ mm d}^{-1}$)	-20 m	-8 m
b_4	-150 m	-150 m

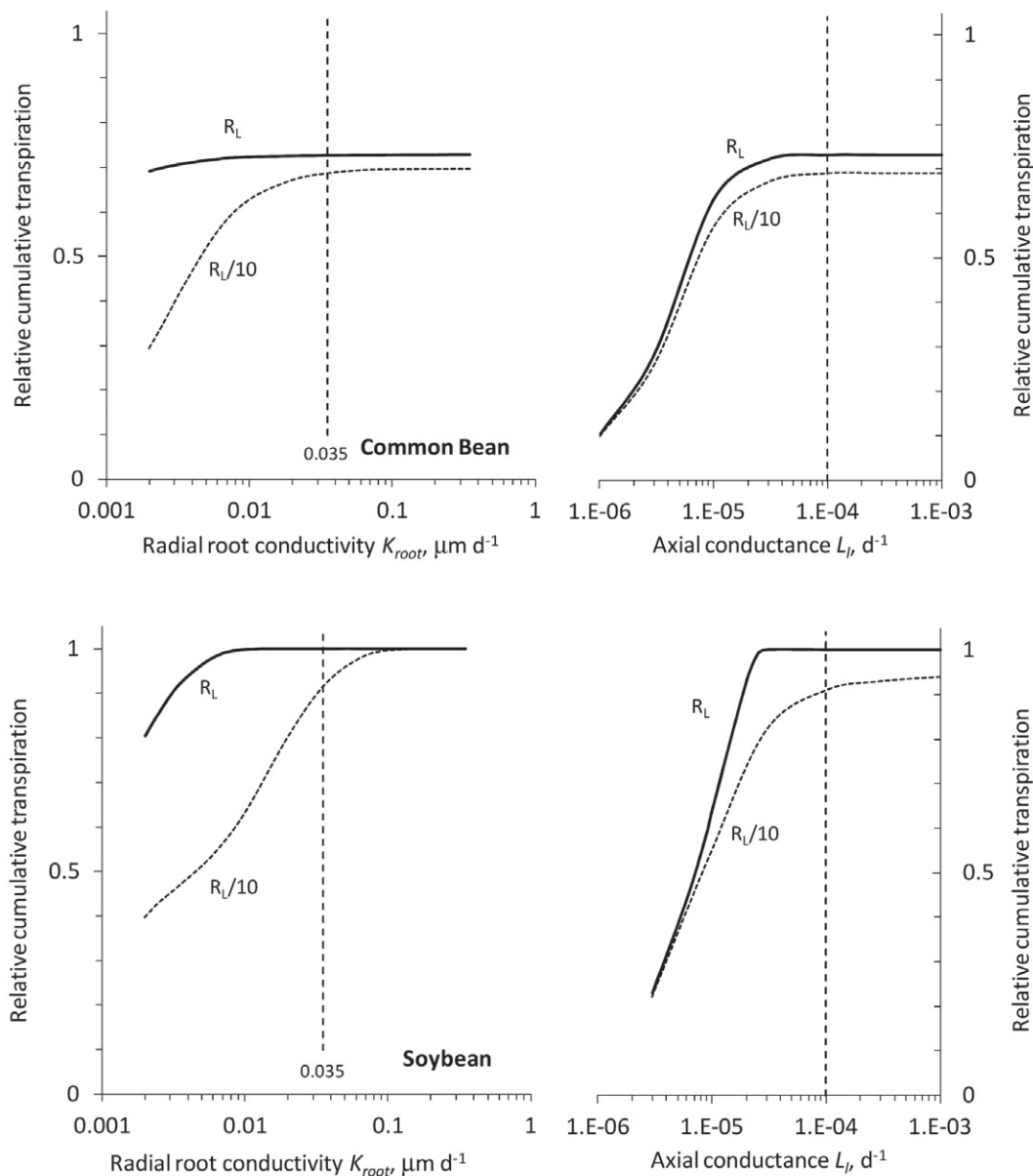


Fig. 10. Predicted relative cumulative transpiration as a function of radial root conductivity and axial conductance for the common bean (*Phaseolus vulgaris* L.) and soybean [*Glycine max* (L.) Merr.] scenario, obtained with the Soil–Water–Atmosphere–Plant (SWAP) model in combination with proposed reduction function (PRF) using the experimentally obtained root length distribution (R_L according to Fig. 3) and with a 10 times thinner root length density ($R_L/10$).

over depth can show important differences (Fig. 11). On this specific date, pressure heads are very negative and soil resistance largely dominates over plant resistance. As a consequence, simulated pressure head profiles for JRF and PRF are similar with a dry layer between 10- and 40-cm depth, but very different from the FRF simulated profile. Moreover, JRF and PRF predict a larger uptake at the bottom of the rooted zone where pressure heads and matric flux potentials are more favorable. This prediction is in agreement with experimental observations about root water uptake distribution by Sharp and Davies (1985) and Novák (1987). In addition, FRF, having simulated more transpiration reduction in the

previous period (Fig. 9) causing pressure heads to be less negative, predicts a more uniform uptake over the entire root zone, except for the first 10 cm where very little is taken up.

The second specific date analyzed in Fig. 11 is 22 January, at the end of a very rainy period within the soybean scenario. The soil is wet, with pressure heads around -1 m, and is even higher in the subsoil. As a consequence, soil hydraulic resistance is very low and plant resistance becomes more important in the PRF simulations. All three models predict uptake to be at potential rates, but uptake profiles FRF and PRF practically coincide, whereas

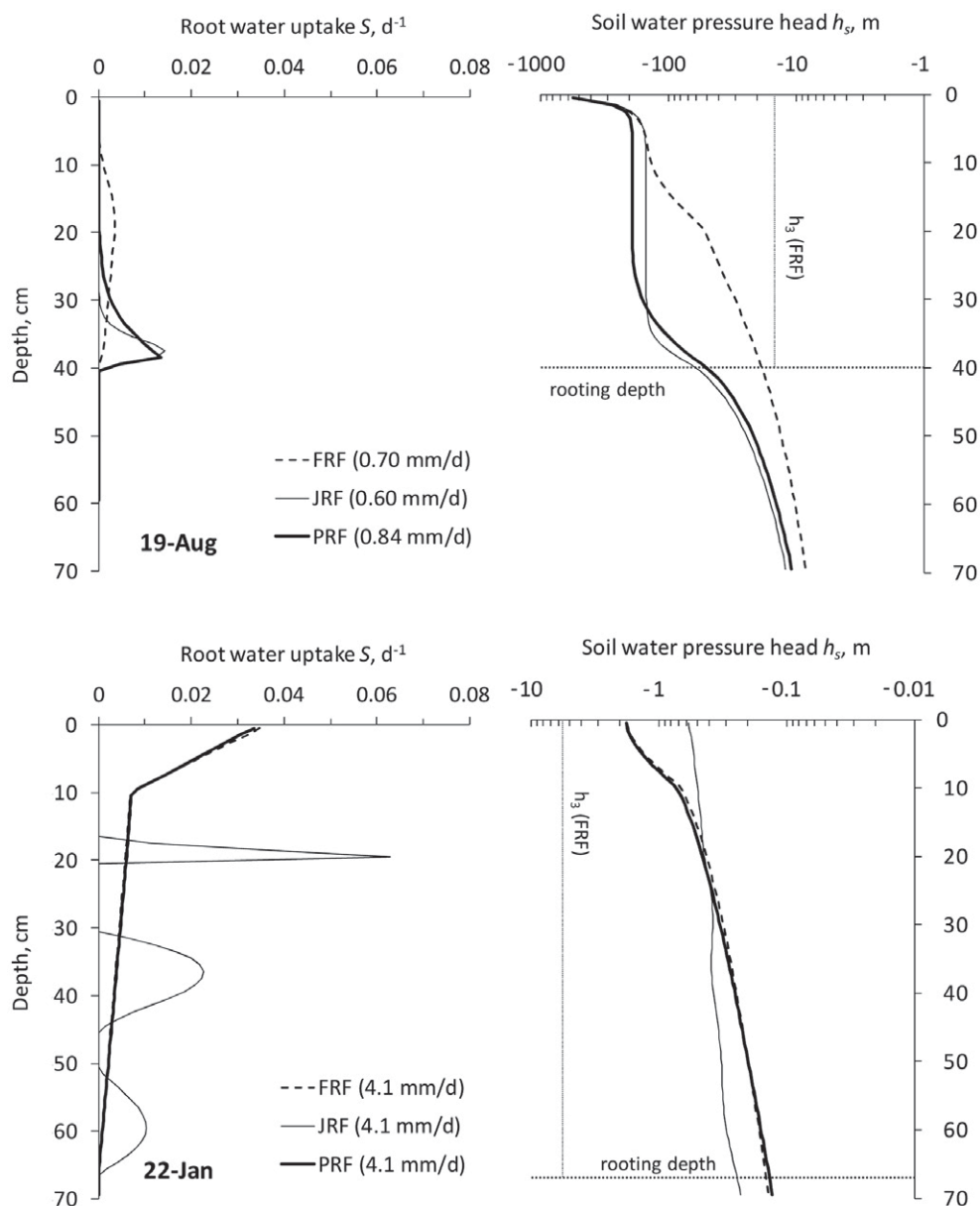


Fig. 11. Simulated water uptake and pressure head profiles on two selected dates, 19 Aug. 2010 during the common bean (*Phaseolus vulgaris* L.) experiment and 22 Jan. 2011 during the soybean [*Glycine max* (L.) Merr.] experiment, for the three reduction functions. [Feddes et al. (1978) reduction function (FRF); de Jong van Lier et al. (2008) reduction function (JRF); proposed reduction function (PRF). Values between brackets refer to total simulated uptake on the respective date].

JRF shows a very different prediction. Under these conditions ($h_s > h_3$), FRF uptake distribution reflects root length density distribution. Similarly, PRF predictions at these low soil resistances are also determined by root length density, very similar to the FRF approach. In contrast, the JRF uptake profile is largely determined by differences in matric flux potential and results in counterintuitive values (Fig. 11). A similar behavior of JRF was described by Durigon et al. (2012).

Statistical indices of comparison root mean square error (RMSE), model efficiency coefficient E (Nash and Sutcliffe, 1970) and R^2

are shown in Table 4 for water content at evaluated depths using the predictions of SWAP with the three transpiration reduction functions and field observations from the common bean and soybean experiments (data represented in Fig. 8). The indices show similar model performance among the three reduction functions. Values of RMSE reflect the average prediction error and are between 0.02 and 0.04 $\text{m}^3 \text{m}^{-3}$. An E less than zero ($E < 0$) indicates that the observed mean is a better predictor than the model, whereas positive values of E show the model to be a better predictor than the observed mean. $E = 1$ corresponds to a perfect match of model predictions and observations. Values of E in

Table 4. Statistical indices of comparison [root mean square error (RMSE), $\text{m}^3 \text{m}^{-3}$, Nash–Sutcliffe model efficiency coefficient E , and R^2] between field observations from experiment with common bean and with soybean and Soil–Water–Atmosphere–Plant (SWAP) simulations of volumetric water content using the three transpiration reduction functions Feddes et al. (1978) reduction function (FRF), the de Jong van Lier et al. (2008) reduction (JRF), and proposed reduction function (PRF).

Depth	Reduction function	RMSE	E	R^2
Common bean				
5 cm	FRF	0.04	0.34	0.606
	JRF	0.04	0.36	0.616
	PRF	0.04	0.32	0.603
15 cm	FRF	0.02	0.71	0.738
	JRF	0.02	0.62	0.722
	PRF	0.02	0.53	0.713
30 cm	FRF	0.04	0.64	0.879
	JRF	0.04	0.32	0.808
	PRF	0.04	0.39	0.853
Soybean				
15 cm	FRF	0.04	−1.58	0.391
	JRF	0.03	−0.49	0.351
	PRF	0.04	−1.90	0.342
30 cm	FRF	0.03	0.27	0.446
	JRF	0.02	0.37	0.470
	PRF	0.03	0.06	0.439

Table 4 are all positive, except for the soybean simulations at 15-cm depth. Values in the soybean simulations may have been affected by the extrapolated values of θ_s and K_s , as previously discussed. Furthermore, due to the very high rainfall frequency (see Fig. 9c), the variation in measured water content over time was very low, making the average value a reasonable estimator, and eventually better than the complex model with any of the reduction functions. The weak correlation between predictions and observations in this scenario is also shown by the low values of R^2 .

The similarity between statistical indices for the three reduction functions shows that the experimental data on water content are not sufficiently informative to critically test differences between the models. More detailed experimental data would be necessary to statistically expose differences, as revealed for example by simulated uptake over depth (Fig. 11).

Concluding Remarks

In this paper, we developed a theoretical framework and associated algorithm to predict water flow and potentials in the soil–root–shoot pathway, and increased insight into the mechanism of root water uptake. The resulting model requires input of root length density, root radial conductivity, and xylem axial conductance, which

makes it less likely to become popular among hydrologists. The examples of application, however, indicate that an understanding of the underlying mechanisms may be used to estimate empirical parameters for simpler models, e.g., the Feddes et al. (1978) reduction function or the available water concept presented by Doorenbos and Kassam (1986). By estimating the pressure head at the root surface as a function of leaf pressure head and bulk soil hydraulic conditions, this method allows computation of the water uptake rate per soil layer. Integrating over an entire root system, we can then estimate the total plant transpiration and transpiration reduction.

We may then conclude that:

1. When using Brooks–Corey soil hydraulic functions, an analytical solution for the developed physically-based model for the simulation of potentials and flows in the soil–root–shoot system was derived for the specific case of $l = 1/3$.
2. Maximum root water uptake rate exists as a function of soil water status, soil hydraulic properties, root length density, and root radius.
3. The model was successfully introduced into the SWAP ecohydrological model as a new transpiration reduction function for layered soil profiles and root length density variations over depth.
4. A step function for the relation between leaf water pressure head and relative transpiration can be used instead of a smooth relation that requires additional empirical parameters.
5. Simulations show root water uptake is not sensitive to radial root conductivity and axial conductance when root length density is high.
6. Under wet conditions (low soil hydraulic resistance), the new model predicts water uptake distribution similar to the Feddes reduction function, reflecting root length density distribution. Under dry conditions (high soil hydraulic resistance), the new model makes predictions of water uptake similar to an earlier presented model which does not include internal plant resistance.
7. The comparison of the new model, the Feddes function, and an earlier presented model which does not include internal plant resistance to data from two field experiments showed similarity between statistical indices for the three models, but did not allow for critical testing of differences between the models.

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Handelsregister: Amtsgericht Bonn, HRB 5919 | Sitz der Gesellschaft Bonn | USt-IdNr.DE 122265872 | WEEE-Reg.-Nr. DE60800328

* Im Inland kostenfrei. Im Ausland gilt die kostenpflichtige Rufnummer +49 171 252 2828 (in Ländergruppe 1 mit Option "Weltweit" kostenfrei).

